AN ECLECTIC TECHNOLOGIES FOUNDATIONAL DEFINITION

# FD-BL — The Bright Line

Foundational Definition: The Placement Criterion, Its Granularity, Its Scope of Application, and the Obligation It Creates

Version 0.6 — DRAFT · 15 October 2026

Tim Roxey

Eclectic Technologies

tim.roxey@gmail.com · 410.474.9240 (mobile/Signal)

SUGGESTED CITATION

T. Roxey, "FD-BL — The Bright Line," Version 0.6, in *FD — Foundational Definitions, November 2026 Edition*, Eclectic Technologies, October 2026. https://doi.org/10.5281/zenodo.23401272

*This version is a member of FD — Foundational Definitions, November 2026 Edition (https://doi.org/10.5281/zenodo.23401272). The edition is a dated coherence claim: it states the family's state as of 19 October 2026 and is not falsified by a later revision. The earlier October 2026 Edition (https://doi.org/10.5281/zenodo.23117612), September 2026 Edition (https://doi.org/10.5281/zenodo.23000686) and August 2026 Edition (https://doi.org/10.5281/zenodo.22181029) remain citable for what they contain. Version state is authoritative in FD-REG, whose dated copy accompanies this edition. All editions: https://doi.org/10.5281/zenodo.22181028.*

---

**Designation:** FD-BL
**Family:** Foundational Definitions
**Version:** 0.6 — DRAFT (issued in the November 2026 Edition; supersedes v0.5)
**Date:** 15 October 2026
**Author:** Tim Roxey
**Organization:** Eclectic Technologies
**Audience:** Standards committee reviewers; regulatory agency technical staff; engineering staff at regulated utilities and industrial facilities; suppliers of ICS/OT products incorporating machine learning or generative AI
**Parent:** None. This instrument defines. See §2.
**Companion root definitions:** FD-LD (layer decomposition and determinism); FD-EV (the envelope); FD-BR (the Broker role-type, of which the Command Broker at §4.5 is this Line's instantiation); FD-LS (local sovereignty)

---

## 1. Purpose and the Defect This Instrument Corrects

The Bright Line is the central concept of this corpus. Roughly forty published instruments inherit it. Sector frameworks, facility class documents, vendor conformance obligations, and regulatory annexes all resolve to it.

At this instrument's first issue it was stated five different ways, and the five were not compatible.

They were not five competing answers to one question. They were answers to four different questions, each mistaken for the whole. A reader arriving at any one instrument acquired a definition that would not survive contact with the next, and a determination made against one formulation did not bind the others. The restatements are conformed at each carrying instrument's next revision, and §7 names those that still vary.

The underlying cause is worth naming, because it governs how this instrument is written. The corpus never stated what *kind of thing* the Bright Line is. Each formulation rests on a different unstated answer to that question — a property of components, a boundary between architectural layers, a threshold on actions — and each is internally coherent given its own unstated premise. The premises were never written down, so they were never examined.

This instrument separates the four questions, answers each once, states its own ontological commitment before defining anything, and establishes which document owns the answer.

---

## 2. Authority, Family, and Convention

### 2.1 Precedence

This instrument owns the definition of the Bright Line for all documents in the corpus.

Foundational Definitions take precedence over sector frameworks, facility class documents, application guides, vendor companions, annexes, and implementation guidance. A sector framework defines *within* a concept; it does not define the concept. This is a tier rule and applies to every foundational definition and every sector framework, not only to the instruments named in this document.

Every other statement of the Bright Line is a restatement. A restatement may be expressed in sector vocabulary and may apply the definition to sector-specific actions. It shall not vary the placement criterion, the granularity, the scope of application, or the obligation. Where an existing instrument varies, this instrument governs and the varying text shall be conformed at that instrument's next revision.

Where an instrument propagates the definition to a series or to subordinate documents, it shall propagate by reference to this instrument rather than by restating text. An instrument that rebroadcasts its own wording will re-introduce variance at every revision cycle.

### 2.2 The Foundational Definitions family

FD-BL is a member of the Foundational Definitions family. A concept belongs in this family when the correctness of other documents depends on agreeing what kind of thing it is. A concept with a settled external meaning, on which no document's correctness turns, does not belong here however important it is.

FD-LD, covering layer decomposition and determinism, is a companion root definition and is the source text for the scope of application at §4.4. FD-LD owns the determinism and layer vocabulary; it does not own placement, which is defined here.

### 2.3 Declaration convention

Each instrument in this family shall state what kind of thing its concept is before defining it. Placement of that statement is at the head of the definitional section.

This convention exists because its absence is the identified cause of the defect at §1. A document whose subject is components will tend to state a concept as a property of components without noticing it has done so. Requiring the declaration makes the commitment visible and therefore reviewable.

### 2.4 What this instrument does not alter

Nothing here alters the criticality band model, the consequence-severity categories, the Command Broker qualification requirements, or the assessment methodology. Those are defined elsewhere and are inherited unchanged.

This instrument is definitional. It states requirements about the definition and about conformance to it. It also states the architectural form of the obligation, at §4.5, and that statement is normative: what a control system shall be incapable of executing, and what its enforcement mechanism shall not be defeasible by. It states no operational procedure for the systems it describes.

---

## 3. The Four Questions

Precision requires that these be kept apart. Nearly every defect in the current text is a collapse of one into another.

**Placement.** On what property is the line drawn — what makes an action fall above it rather than below it?

**Granularity.** At what unit is placement determined — the system, the application, or the individual action?

**Scope.** Which components are subject to Bright Line analysis at all?

**Obligation.** What does an above-line placement require?

Placement and scope are the pair most often conflated, and that conflation produces the most consequential errors in the current corpus. The determinism property answers the scope question. It does not answer the placement question. A framework that uses determinism to decide where the line falls has made a component's technology the measure of a plant's consequence, which is the reasoning this corpus exists to displace.

---

## 4. The Definition

### 4.1 What kind of thing the Bright Line is

The Bright Line is a threshold on actions.

It is not a property of components. It is not a boundary between architectural layers. It is not a classification of technologies. It is a threshold, and what it partitions is the set of actions a system can take.

Components enter the analysis in one respect only: whether they meet the determinism carve-out decides whether the actions they produce are subject to the threshold. That is the scope question at §4.4, and it is a separate question from where the threshold falls.

Everything else in this instrument follows from this paragraph.

### 4.2 The placement criterion is consequence

The Bright Line is drawn on the consequence of the action. It is not drawn on the technology that produced the action, on the sophistication of that technology, on whether the action is characterized as advisory or as control, or on the architectural layer in which the producing component resides.

An action falls above the Bright Line when its incorrect performance could produce a consequence at or above the severity threshold established for the facility. It falls below when it could not. The consequence is that of the change the action itself would make, assessed over its whole trajectory and rate (§4.3). A consequence that would arise only through an authorization another party grants on the strength of the action's output is not the action's own consequence; it is governed by FD-BL-D2.

Three corollaries follow, and each corrects an error presently in circulation.

A supplier's characterization of a product does not determine placement. A component described as advisory, as decision support, as optimization, or as operator-free autonomy is placed by what its actions can cause, not by what it is called.

Placement does not follow architectural layer. An action originating in a component described as advisory may be above the line if its consequence is above the threshold. An action originating in a control-layer component may be below the line if its consequence is not. The advisory-to-command seam is a useful place to build enforcement; it is not the definition of the boundary. Building a single gate there on the assumption that it coincides with the line will leave above-line actions ungated on the advisory side and gate below-line actions unnecessarily on the command side.

Placement does not follow determinism. Whether a component is deterministic governs whether it is subject to this analysis, which is §4.4. It does not govern where the line falls.

**Output that makes no change of its own.** Output of an in-scope component that would make none of the changes listed at §4.3, even if acted upon, is placed below the line as a process action, and is routed and recorded as such. *Placed as a process action* means placed on the changes the output would itself make, of which there are none. Where it can function as the basis for an above-line authorization, or as an input on which placement or discharge depends, that role does not move it above the line; it attracts the indication-integrity obligation of FD-BL-D2. Output that, once acted upon, would make one of the listed changes, such as a work order that takes equipment out of service, is an action under §4.3 and is placed on its consequence.

### 4.3 Placement is determined action by action

The unit of placement is the individual action, not the system and not the application.

**What one action is.** An action is one change that a component's output would make, if executed or acted upon, to a process parameter; to a protective function's settings, trip logic, bypass state or actuation criteria; or to the pre-authorized envelope or the gatekeeper. An action is individuated by the change it makes, not by the message that carries it or by the objective of the component that proposed it. Its consequence is assessed over its whole trajectory and rate.

Three rules close the definition. A recommendation and its acceptance are one action: the acceptance is the authorization of that action, not a second action of the component. A change commanded as one profile is one action. No action may be individuated more finely than the change it makes in order to lower its placement; where the parts would place lower than the whole, the whole is placed. Composition of actions that are properly separate is governed by FD-BL-D1 R2 wherever advance discharge applies, and is not addressed here.

A release a vendor makes to a population of hosts outside any one facility is not, by itself, an action under this section. Its controls on the vendor's side are obligations the asset owner-operator places on the vendor by contract, derived from the facility's design basis. At the facility, taking in a release that would change a control or protective function is an action of the kind this section defines, placed on its consequence, and it comes in only through a governed act (*Local Sovereignty* §3.2).

Where a component within scope releases a change in stages, each widening is its own action. The first release is placed on the consequence of its whole planned reach, not on its first stage alone.

A change that a component within scope makes to an enterprise network, including isolating a segment, pushing a reimaging or shutting down a service, is an action under this section and is placed on its consequence, which may reach functions the network carries for others.

The definition is for this Line. FD-BR §4.4 leaves the unit to each Line document, and this paragraph sets it for the industrial Line only.

A single product, deployed once, in one facility, will typically produce actions on both sides of the line. A generative advisory system that drafts maintenance work orders, recommends setpoint adjustments, and summarizes historian data produces actions, and output that makes no change of its own (§4.2), whose consequences differ by orders of magnitude. Classifying the product as a whole is imprecise in both directions: it imposes interposition on trivial actions, which produces workarounds, and it obscures which specific actions actually require it.

The practical consequence for suppliers is that the required architecture is a consequence classifier that routes action by action, not a single gate at a seam. The classifier's routing decisions, and the basis on which each action was classified, are part of the evidentiary record.

Placement is distinct from criticality band assignment, and the two operate at different units. Band assignment is a property of the application, derived from its consequence-severity profile, and establishes the rigor of assessment, qualification, and audit that applies. Placement is a property of each action within that application and, for an in-scope component, establishes whether the action may proceed without human authorization. A high-band application contains below-line actions. A low-band application may contain above-line ones. Neither determines the other, and an instrument that treats band assignment as placement has answered the granularity question wrongly.

### 4.4 Scope of application: the determinism carve-out

Bright Line analysis and the Command Broker obligation apply to every component that does not meet the determinism carve-out. FD-LD §4.3 states that scope rule as governing, together with the carve-out's property test, its justification, the prohibition on narrowing it to enumerated roles, the guard that it is not a grant of ungoverned autonomy, and the ecosystems in which obligations on carved-out components arise. They are not restated here. This section carries what is specific to this Line: that the carve-out is not the line, and that it does not extend to actions taken upon carved-out components.

**The carve-out is not the line.** Determinism decides whether a component is subject to Bright Line analysis. Consequence decides where the line falls for the actions that component produces. A formulation that places deterministic machine learning below the line and generative systems above it has fused the two questions and produces two errors at once: it implies that every generative action is above the line, including those with no material consequence, and it implies that a deterministic component can never produce an above-line action, which is false and which removes from analysis the automated systems most capable of causing harm.

**The gaps are recorded in FD-LD.** The coverage gap for deterministic machine learning taking autonomous, consequential control action, and the assessment-method gap for in-scope components that are not generative, are recorded at FD-LD §4.8, so that the carve-out is not read as an assertion that no gap exists.

**The carve-out does not extend to actions taken upon carved-out components.** A component within scope of this instrument that modifies a protective function's settings, trip logic, bypass state, or actuation criteria is taking an action upon that function. Such actions are placed on their own consequence under §4.2, and they are among the highest-consequence actions available in any facility. The carve-out covers a carved-out component's own actions. It does not cover in-scope components acting upon it. Where the component that modifies them is itself carved out, the case lies in the coverage gap recorded at FD-LD §4.8, which this instrument does not resolve.

### 4.5 The obligation above the line

An action placed above the Bright Line requires human authorization that is substantive and attributable. FD-BR §4.5 defines both terms, and they are not restated here.

The authorizing human is the Command Broker, who discharges the obligation in the moment or, where FD-BL-D1 applies, in advance as the author of the pre-authorized envelope (FD-BR §4.1, §4.2). FD-BR is the source of the Broker role-type and of the pairing of each Line with exactly one Broker (FD-BR §4.1, §4.2); this section carries the domain-specific content of the role for this Line. Broker competency, onboarding, and accountability are established in the qualification standard and are not restated here.

The obligation is architectural, not procedural. The enforcement mechanism is the Gate of FD-BR §4.1. The control system shall be incapable of executing an above-line action originating from a component within scope of §4.4 without the required authorization. The enforcement mechanism shall not be defeasible by configuration change, software update, or system integration, and the requirement is not discharged by a procedure instructing an operator to confirm.

Accountability for the authorized action rests with the named human broker and with the asset owner-operator, the industrial instance of FD-BR's deploying authority (FD-BR §4.5). A supplier may build and operate the enforcement mechanism; a supplier may not hold the accountability, and no software component holds it.

**Why the obligation attaches where it does.** The Command Broker is compensation for unverifiability. It is not a substitute for automation, and it is not a general requirement that consequential actions have human involvement.

This explains an asymmetry that would otherwise be arbitrary. A deterministic protective function — reactor trip, engineered safety feature actuation, protective relay operation, automatic load shedding — performs actions of the highest consequence without a Command Broker. Those actions are not below the line; asserting that they were would be false, and would break the criterion at §4.2. They are outside the obligation under the carve-out at §4.4, and the reason the carve-out is defensible is that a protective function's behavior is bounded by proof. What it will do is known before it does it. Judgment is not required where proof is available.

Where behavior cannot be bounded by proof, the corpus requires a qualified human to supply the judgment that proof cannot. That is the whole of the Command Broker's purpose, and it is the criterion against which any proposed alternative discharge of the obligation is to be evaluated.

**Substitutability of the Command Broker.** FD-BR §4.4 requires each Line document to state its own substitutability value, and owns the rule that sets it. This Line's value is that **the Command Broker is partially substitutable.** For a narrow and specified class of actions, the Command Broker may discharge the authorization in advance and a verified gatekeeper then admits each action in the class, but only under R1 to R3 of FD-BL-D1. What the gatekeeper substitutes for is the Broker's presence at the moment of action, not the Broker's judgment (FD-BR §4.4). An action that changes the pre-authorized envelope, or the gatekeeper's configuration, limits or logic, is excluded from advance discharge under any construction (FD-BL-D1 R3). The value is determined at FD-BL-D1, recorded at §6.1, and stated here by pointer, not restated.

**The unit of authorization is the individual action**, placed under §4.3. An above-line action is authorized one at a time by the Command Broker. Where FD-BL-D1 applies, the object a human authorizes in advance is the pre-authorized envelope. It is accepted by the Command Broker discharging in advance, together with the R2 completeness proof, under the advance-mode competence FD-BL-D1 §7 requires. Each action inside the envelope is then admitted by the gatekeeper rather than by the Broker. Mode of discharge varies. Placement, granularity and scope do not.

---

## 5. Worked Examples

*These examples are illustrative and interchangeable. Sector-specific variants may be substituted for a given audience without altering the definition.*

A supervised learning model performing vibration-based bearing failure prediction on a feedwater pump, deterministic within a stated input bound and formally testable as deployed, is outside the scope of Bright Line analysis under §4.4. If the same model is replaced by a generative system producing the same recommendations, the system enters scope, and each of its actions is then placed on consequence: a maintenance work order recommendation is placed on what taking the pump out of service could cause, and in the ordinary case falls below the line; and a recommendation that adjusts pump speed during operation is placed on what an incorrect adjustment could cause.

A generative optimization system on a natural gas processing train nominates operating points within an envelope whose setback from the safety instrumented system trip points was established by human

authority in advance. The nomination of an operating point inside the envelope is one action. An adjustment of the envelope itself toward a trip point is a different action with a materially different consequence, and the two are placed separately. The second is above the line under any reading. The first is the action addressed by the mode-of-discharge determination at §6.1 (FD-BL-D1).

A generative system synthesizes plant state into a situational awareness display. The display changes no process parameter, so it is placed below the line as a process action (§4.2). Its role in grounding authorizations does not move it across the line, but that role is where its danger lies. An operator who authorizes an above-line action against a corrupted or drifted synthesis is not blind; he is confident and wrong, and the human authorization the framework relies on has been hollowed out without any protective layer registering a fault. The display therefore carries the indication-integrity obligation of FD-BL-D2 (§6.2): it must be traceable to independently verifiable values, and it may not be the sole basis for the authorization.

A generative system recommends a change to a protective relay setting. The relay is a carved-out component under §4.4 and its own protective actions are outside the obligation. The recommendation is not. It is an in-scope component acting upon a carved-out component, it is placed on its own consequence, and an incorrect protective setting is among the most consequential outcomes available.

A vendor characterizes a product as providing turnkey autonomous process optimization. The characterization has no effect on placement. Each action the product can take is classified on its consequence, and the actions above the line require the architecture at §4.5 regardless of what the product is marketed as doing.

---

## 6. Determined Items

The three items reserved at this section in prior versions are now determined. Each is carried by a dedicated instrument; the resolutions bind this definition by reference and are not restated here. The constraints this instrument placed on each determination (that a resolution may vary mode, not placement/granularity/scope) were honored by the issued instruments.

### 6.1 Mode of discharge for the authorization obligation — resolved; see FD-BL-D1

Whether the §4.5 authorization may be discharged in advance for a narrow class of actions is determined by **FD-BL-D1 (Mode of Discharge: Pre-Authorization by Verified Gatekeeper)**. Advance discharge is permitted only under R1 (the whole safety case in a deterministic, formally verified gatekeeper the proposer cannot modify or bypass), R2 (completeness and soundness proven over state, trajectory, and rate — bounding selection, not only admissibility), and R3 (envelope changes categorically excluded from advance discharge). The consequence classifier and the verified gatekeeper are two objects in series, not one. The composite-classification and third-tier routes remain foreclosed. Mode varies; placement, granularity, and scope do not.

### 6.2 Indication integrity — resolved; see FD-BL-D2

Whether a below-line placement is available to output of an in-scope component that functions as the basis for an authorization is determined by **FD-BL-D2 (Indication Integrity)**. The disposition is a qualifier, not a placement change: such output must be traceable to independently verifiable values (I2) and must

not be the sole basis for an above-line authorization (I3), with the obligation attaching by function (I1). The rule binds both the human-broker and the verified-gatekeeper cases.

### 6.3 Terminology reservation — resolved; see FD-EV

The two-sense collision on "envelope" is resolved by **FD-EV**, a Foundational Definition companion to this instrument and FD-LD. "Envelope" (and "pre-authorized envelope") is reserved for the authorization sense; the operational sense — a region whose violation is a runtime event to detect and report — is renamed "operating band." Conformance is staged per instrument revision, following the "Bright Line" reservation model.

---

## 7. Conformance

This instrument governs the definition of the Bright Line. Restatements shall be conformed at each instrument's next revision, and the tier rule at §2.1 applies whether or not an instrument is named below.

Instruments presently varying from this definition, and the nature of the variance:

**Command Broker Framework, Cross-Sector** — at v0.3 states placement as a threshold on actions and carries action-level granularity (read 2 October 2026); its definition of an in-scope component as "a generative or otherwise non-deterministic component" is narrower than the scope rule at FD-LD §4.3, and it cites FD-LD §3 and this instrument's §4.4 for scope rather than FD-LD §4.3. Noticed to its owning project on 2 October 2026.

**Layer Decomposition and Determinism Terminology Annex** — superseded by FD-LD, which is issued as a Foundational Definition in this family and is the companion root named at §2.2. The annex mapped the layer boundary onto the placement criterion; that mapping does not survive into FD-LD, and no conformance is owed by the superseded annex.

**DBA-EN Energy Sector Framework** — states placement as a technology-class boundary and propagates that statement series-wide by restatement rather than by reference.

**DBA-ES Sector Framework** — restates the technology-class formulation; narrows the carve-out to enumerated roles.

**DBA-VC Vendor Companion** — conforming as written; requires citation of this instrument as the governing definition.

**ISA Parent Framework, IEEE V&V Cognitive AI Framework, CB-Annex-Nuclear** — state the obligation in terms coupled to §6.1, now determined by FD-BL-D1; conformed in the Pass 2 amendment set.

The detailed diagnosis, replacement text, and sequencing for each instrument are carried in the separate conformance schedule and are not reproduced here.

---

## 8. Author Decisions Outstanding

**One item closed in the October 2026 Edition.** The confirmation of specific standard and clause references at §4.4 is no longer needed. The placeholder is removed, the ecosystems are named in FD-LD §4.3, which owns the scope rule, and no clauses are cited.

**Open:** whether the conformance schedule is issued alongside this instrument or held for release on request.

---

## 9. Foundational References


Roxey, T.E. *Generative AI Risk in Industrial Control Systems: A Study for Critical Infrastructure Operators, Regulators, and Suppliers.* Version 29. Eclectic Technologies, 2026.

Command Broker Framework, Cross-Sector. Roxey, T.E. Eclectic Technologies, 2026.

Layer Decomposition and Determinism: The Two-Layer Architecture, the Determinism Boundary, and the Machine Learning Carve-Out. Roxey, T.E. Eclectic Technologies, 2026. (Superseded by FD-LD.)

DBA-VC Vendor Companion. Roxey, T.E. Eclectic Technologies, 2026.

DBA-EN Energy Sector Framework; DBA-ES Sector Framework; DBA-GAS-FC2 Natural Gas Processing. Roxey, T.E. Eclectic Technologies, 2026.

Foundational Definitions and their determinations are cited here by designation, without version tokens; FD-REG holds their version state. FD-LD — *Layer Decomposition and Determinism*. FD-BR — *The Broker*. FD-EV — *The Envelope*. FD-BL-D1 — *Mode of Discharge: Pre-Authorization by Verified Gatekeeper*. FD-BL-D2 — *Indication Integrity*. Eclectic Technologies, 2026.


---

## Revision History

**0.6 — 15 October 2026.** Issued in the November 2026 Edition, and recorded in FD-REG v0.32 in the same act. §4.3 gains three paragraphs after its closing rules, carrying the text queued in FD-REG v0.28 and v0.30 from two author determinations of 14 October 2026. From Determination_2026-10-14_Three_Author_Decisions_Continuity_Classes_Vendor_Releases_and_Staged_Releases: a vendor's release to a population of hosts outside any one facility is not by itself an action under §4.3, its vendor-side controls being contractual obligations of the asset owner-operator derived from the facility's design basis, while taking in such a release at the facility is an action placed on its consequence and enters only through a governed act; and each widening of a staged release is its own action, the first placed on its whole planned reach, which keeps the third closing rule. From Determination_2026-10-14b_Enterprise_Network_Changes_Count_and_Indication_Integrity_Requires_Sufficiency: a change a component within scope makes to an enterprise network is an action placed on its consequence. The definition of one action is not otherwise changed; placement, scope and the obligation are untouched. No section is renumbered.

**0.5 — 2 October 2026.** Issued in the October 2026 Edition, and recorded in FD-REG v0.22 in the same act. One cumulative entry, extended at each step of the Action Plan for the October 2026 Edition that opens this instrument (Steps 1, 2, 4 and 5).

*Step 1, 2 October 2026 (F-08, F-11; open question (d)8; F-16, F-25, F-26; F-24 checked).* §4.3 defines one action, on the author's determination of 2 October 2026 (Decision A, Option A2): an action is individuated by the change it makes, its consequence is assessed over its whole trajectory and rate, and three rules close the definition. Composition of separate actions stays with FD-BL-D1 R2, which is pointed to and not restated, and the definition is stated to be for this Line. This defines the unit of granularity, which earlier versions named without defining; it does not change placement, scope or obligation. §2.4 now states that the architectural form of the obligation at §4.5 is normative and that the instrument states no operational procedure. The §4.3 sentence on what placement establishes is confined to in-scope components. §1 restates the five-formulations claim in the past frame. §4.5 cites FD-BR §4.5 for "substantive" and "attributable" instead of restating them. §7 names FD-LD as issued and the Layer Decomposition annex as superseded. The render sentence in the v0.4 entry below is qualified as referring to the August 2026 Edition. §§1–5 were checked for provenance narrative under F-24 and carry none to move, so no Record section is added at this step. §4.4, the sentence of §4.5 naming the authorizing human, the verbs of advance discharge, §4.2 and §5 are untouched; they belong to Steps 2, 4 and 5.

*Step 2, 2 October 2026 (F-02 and F-20 on FD-BL's side; F-25).* §4.4 now cites FD-LD §4.3 for the scope rule, stated there as the complement of the carve-out (Decision B, Option B3), and no longer restates it. Its paragraphs on the carve-out's justification, its property test, the prohibition on narrowing it to enumerated roles, and the not-a-grant-of-autonomy guard with the ecosystems named under it move to FD-LD §4.3. Its gap paragraph moves to FD-LD §4.8 and is replaced by a pointer (F-20). The paragraphs on the carve-out not being the line and on actions taken upon carved-out components stay, the second now saying "settings" to match §4.3 and adding that a carved-out actor lies in the gap FD-LD records. The placeholder on standard references is removed and the first item in §8 is closed (F-25).

*Step 4, 2 October 2026 (F-03 and F-05 on FD-BL's side; F-26).* §4.5's first sentence now states that the Command Broker discharges in the moment or, where FD-BL-D1 applies, in advance as the author of the pre-authorized envelope (Decision C, Option C1). The substitutability paragraph aligns the verb, the Command Broker discharging in advance and the verified gatekeeper admitting, and states that the gatekeeper substitutes for the Broker's presence and not its judgment. The unit-of-authorization paragraph names the Command Broker in place of "a qualified envelope author." §4.5 names its enforcement mechanism as the Gate of FD-BR §4.1 (F-05), and maps the asset owner-operator as the industrial instance of FD-BR's deploying authority (F-26).

*Step 5, 2 October 2026 (F-04 on FD-BL's side; F-13; Finding S1-1).* On the author's determination of 2 October 2026 (Decision D, Option D1, with the general form decided the same day as a supplement), §4.2 gains a paragraph placing output that would make none of the changes listed at §4.3, even if acted upon, below the line as a process action, routed and recorded as such, with the FD-BL-D2 obligation attaching where it can ground an above-line authorization or feed placement or discharge. This is the qualified form of the sentence FD-BL-D2 relied on, now in the definition so that FD-BL-D2 cites a definition and not an example (F-13). It closes Finding S1-1. The §5 situational-awareness example places the display below the line as a process action and states the FD-BL-D2 obligation it carries, so that FD-BL and FD-BL-D2 say one thing (F-04). §6.2 speaks of the output of an in-scope component rather than generative output, following Decision B.

*Verification read, 2 October 2026.* §4.1 says that whether a component meets the determinism carve-out, rather than its determinism alone, decides whether its actions are subject to the threshold. §4.3's illustration says the advisory system produces actions and output that makes no change of its own, matching §4.2. §7's line on the Command Broker Framework, Cross-Sector is corrected: at its v0.3 it

states placement on actions and carries action-level granularity, and its in-scope definition and its citation for scope are now the variance, which was noticed to its owning project on 2 October 2026. No change to placement, granularity, scope or obligation.

*Review of 2 October 2026.* On the author's instruction to apply the cold review of the October 2026 Edition review draft as recommended (FD-BL Reconciliation/Independent_Review_2026-10-02_FD_October_2026_Edition_Review_Draft_v0_1), this instrument takes findings G-02, G-03, G-04, G-14, G-17; each edit is quoted in the review's application record.

**0.4 — 27 September 2026.** Conforms §4.5 in two respects, both found on 27 September 2026 in preparing FD-BR v0.7. First, the *Open at this version* paragraph, which described advance discharge as a determination in progress, is replaced. §6.1 had recorded that question as resolved by FD-BL-D1 since v0.3, so the section contradicted the one it pointed to. In its place, §4.5 states this Line's substitutability value by pointer, as FD-BR §4.4 requires of each Line document: the Command Broker is partially substitutable, under FD-BL-D1 R1 to R3, with envelope changes excluded under R3. It also states the unit of authorization: the individual action, and the pre-authorized envelope accepted by a qualified author where FD-BL-D1 applies. Second, §4.5 cites FD-BR as the source of the role-type and the pairing invariant, which FD-BR §7 had asked of it. The header names FD-BR as a companion root. §9 adds the family's instruments by designation, without tokens. **No change to placement, granularity, scope or obligation**, and §§1–4.4, §5, §6, §7 and §8 are unchanged. FD-BL-D1 predates FD-BR and does not cite it. FD-BL-D1 is not revised in this act, because the pointer from this Line is enough to carry the rule-and-value arrangement. When this entry was written, the August 2026 Edition render 01_FD-BL_The_Bright_Line_v0_3 was one version behind this master. That render is a member of the August edition and stays as published there; the September 2026 Edition carries v0.4.

**0.3 — August 2026.** §6 open items collapsed to resolved pointers, all three now determined: §6.1 mode of discharge → FD-BL-D1 (R1–R3; two-object model; foreclosures retained); §6.2 indication integrity → FD-BL-D2 (I1–I3; human and machine cases unified); §6.3 terminology reservation → FD-EV (envelope reserved for the authorization sense, operational sense renamed "operating band"). Worked-example cross-references and the §7 conformance line updated to cite the issued determinations. No change to the definition (§§1–5).

**0.2 — August 2026.** Designated FD-BL and placed in the Foundational Definitions family. §2 restructured to carry the tier rule (foundational definitions take precedence over sector frameworks generally), the family membership test, the propagate-by-reference requirement, and the declaration convention. New §4.1 states what kind of thing the Bright Line is, in compliance with that convention; subsequent subsections renumbered. §4.4 carve-out: role-list narrowing struck, property test retained, scope statement added recording that the carve-out is not a grant of ungoverned autonomy and naming where obligations arise, with the coverage gap for machine-learned autonomous control acknowledged rather than papered; new provision that the carve-out does not extend to in-scope components acting upon carved-out components. §4.5 adds the verification rationale — the Command Broker is compensation for unverifiability — and derives the protective-function asymmetry from it. §6.1 constraint retained and its foreclosures named explicitly, with reasons and with the override of the nuclear annex owner's stated preference recorded. §7 reduced to an operative register; diagnosis and sequencing moved to a separate conformance schedule. Worked example added for actions taken upon carved-out components.

**0.1 — August 2026.** Initial draft. Four-question frame; canonical definition; carve-out distinguished from the line; full reconciliation register.